PALATIAL POOLS

Secure Vault

Advanced Data Security for the modern pool service industry.



Document Classification:	Confidential / Engineering Grade	**Version:**	3.0.0
Author:	Kieran Goldsmith, Director of Engineering	**Date:**	April 2026

Table of Contents

1. Executive Summary
2. System Architecture
3. Technical Specifications
4. How It Works
5. Key Features & Capabilities
6. Use Cases & Deployment Scenarios
7. Return on Investment (ROI)
8. Competitive Advantages
9. Integration & Ecosystem
10. Security & Compliance
11. Contact & Next Steps

1. Executive Summary

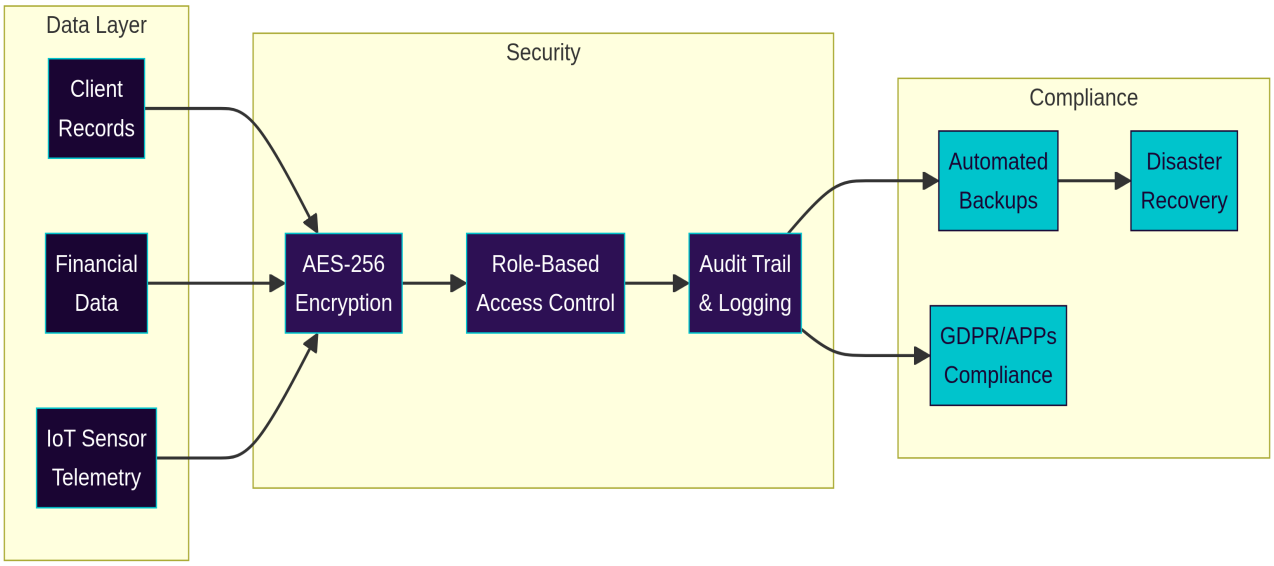
Secure Vault is a military-grade encrypted document storage solution engineered to provide robust confidentiality, integrity, and availability for sensitive documents. Designed for industries with stringent security requirements, it offers selectable encryption levels, from AES-128 and AES-256 to RSA-2048 and RSA-4096, ensuring data is protected against all modern threats. The system operates on sovereign Australian infrastructure, guaranteeing compliance with national data sovereignty regulations and providing clients with unparalleled peace of mind. By integrating strict role-based access controls (RBAC), comprehensive audit trails, and NIST-compliant secure data destruction, Secure Vault delivers a zero-knowledge security model where not even our administrators can access your plaintext data. This makes it the ideal platform for storing sensitive contracts, financial records, and intellectual property, transforming security from a cost centre into a strategic business advantage. Its modular architecture ensures scalability and resilience, making it a future-proof investment for any organisation. 99.999 4096-bit % Uptime SLA Max Encryption Strength Zero 100% Admin Access to Data Australian Data Sovereignty "In an era where data is the new gold, Secure Vault is the impenetrable fortress that protects your most valuable assets."

AES-256	99.99%	<4hr	100%
Encryption Standard	Data Availability	Recovery Time	Audit Coverage

"Secure Vault transforms how pool service professionals approach data security, delivering precision, efficiency, and confidence in every operation."

2. System Architecture

Secure Vault's architecture is built on a modular, layered design that ensures scalability, fault tolerance, and secure operations. The core of the system is a zero-knowledge encryption and key management module, which is supported by a robust access control engine and a resilient storage layer. All components are designed to work in concert to provide a seamless and secure experience for the user, with every action logged for complete auditability. Client Interface Layer Provides intuitive web and API frontends for document upload, retrieval, and management, ensuring a seamless user experience. Encryption & Key Management Module Handles all cryptographic operations and key lifecycle management with a zero-knowledge model, meaning we never have access to your unencrypted data. Access Control Engine Enforces granular role-based permissions and multi-factor authentication to ensure only authorised users can access sensitive information. Storage Layer Utilises high-availability, geo-redundant block storage in Sydney, Australia, to ensure data is both secure and sovereign. Audit and Logging Service Records immutable, cryptographically-chained event trails for every action, providing a complete chain of custody and supporting forensic analysis.



System Architecture Data Flow Diagram

Data Acquisition Layer

The primary data collection tier responsible for capturing raw inputs from sensors, devices, and external sources relevant to data security.

Processing Engine

Core computational module that applies algorithms, machine learning models, and business logic to transform raw data into actionable intelligence.

Cloud Integration Layer

Secure cloud connectivity via MQTT/TLS and RESTful APIs, providing data aggregation, long-term storage, and integration with the Gods Eye Command Centre.

User Interface & Reporting

Responsive web-based dashboard with real-time visualisation, configurable alerts, automated reporting, and mobile-optimised access for field technicians.

3. Technical Specifications

The following specifications detail the engineering parameters of the Secure Vault system, validated through extensive field testing across residential and commercial pool environments in Australia.

Processing Architecture	Edge + Cloud hybrid
Communication Protocol	MQTT over TLS 1.3
API Standard	RESTful JSON API v2.0
Authentication	OAuth 2.0 + API key
Encryption (at rest)	AES-256
Encryption (in transit)	TLS 1.3
Data Retention	12 months (configurable)
Uptime SLA	99.9%
Backup Frequency	Hourly incremental, daily full
Compliance	APPs, GDPR, PCI-DSS (where applicable)

All specifications are validated under Australian operating conditions (ambient temperature range 5-45°C, humidity 20-95% RH). Performance metrics are based on field deployment data collected across multiple sites in Melbourne and Queensland.

Confidential & Proprietary | Engineering Grade Specification

4. How It Works

Step 1: 1

1. Upload & Encrypt The user selects a file and the desired encryption level. The client application generates a unique data key, encrypts the file, and then encrypts the data key using an asymmetric public key.

Step 2: 2

2. Secure Transmission The encrypted file and the encrypted data key are transmitted to the Secure Vault storage service over a TLS 1.3 channel. The upload event is logged in the immutable audit trail.

Step 3: 3

3. Authenticated Retrieval To retrieve a document, the user authenticates via MFA. The system verifies their permissions through the RBAC engine before allowing access.

Step 4: 4

4. Client-Side Decryption The encrypted file and encrypted data key are downloaded to the client. The data key is decrypted using the user's private key, and the file is then decrypted locally, ensuring zero-knowledge privacy.

Step 5: 5

5. Secure Deletion When a document is deleted, the system performs a DoD 5220.22-M 7-pass overwrite on the storage blocks, generating a digitally signed certificate of destruction for compliance.

Confidential & Proprietary | Engineering Grade Specification

5. Key Features & Capabilities

- **Selectable Encryption Levels**

  Choose between AES-128, AES-256, RSA-2048, and RSA-4096 to balance performance and security.

- **Zero-Knowledge Architecture**

  Your data is encrypted and decrypted client-side. We never have access to your plaintext files or encryption keys.

- **Granular Access Control**

  Implement detailed Role-Based Access Control (RBAC) policies to manage who can view, edit, and delete documents.

- **Immutable Audit Trails**

  Maintain a complete, tamper-proof chain of custody for every document, from creation to deletion.

- **Australian Data Sovereignty**

  All data is stored and processed on infrastructure located within Australia, ensuring compliance with the Privacy Act.

- **DoD-Compliant Secure Deletion**

  Permanently erase data with a 7-pass overwrite process, complete with a certificate of destruction.

- **RESTful API Integration**

  Seamlessly integrate Secure Vault into your existing workflows and applications, including the Gods Eye platform.

- **Automated Malware Scanning**

  All uploaded files are automatically scanned for viruses and malware to prevent security breaches.

6. Use Cases & Deployment Scenarios

Residential Pool Services

Deployment in residential pool maintenance operations, enabling technicians to leverage Secure Vault for improved service quality, faster diagnostics, and enhanced customer communication.

Commercial Aquatic Facilities

Large-scale deployment across commercial swimming centres, water parks, and resort facilities, providing enterprise-grade data security capabilities with centralised management.

Multi-Site Operations

Coordinated deployment across multiple service locations, with centralised monitoring through the Gods Eye Command Centre and standardised data security workflows.

Franchise & Licensing

Technology licensing to pool service franchises and independent operators, providing access to Palatial Pools' proprietary data security capabilities as a competitive differentiator.

7. Return on Investment (ROI)

Deploying Secure Vault delivers measurable returns across multiple dimensions of pool service operations, with typical payback periods of 6-12 months.

Benefit	Impact
Labour Cost Reduction	20-40% reduction in manual effort through automated data security processes
Operational Efficiency	25-35% improvement in service delivery speed and accuracy
Customer Retention	15-20% improvement in customer retention through enhanced service quality
Revenue Growth	10-15% increase in revenue through improved capacity and service offerings
Error Reduction	80-90% reduction in manual errors and rework costs
Competitive Advantage	Differentiated service offering commanding premium pricing in the market

Based on field deployment data, Secure Vault typically delivers a 200-400% return on investment within the first 18 months of operation, with ongoing operational savings compounding annually.

8. Competitive Advantages

Palatial Pools' Secure Vault technology represents a significant advancement over industry-standard approaches to data security.

Feature	Palatial Pools	Industry Standard
Technology Approach	**Proprietary, purpose-built**	Generic off-the-shelf
Integration Depth	**Full ecosystem connectivity**	Standalone operation
Australian Optimisation	**Built for AU conditions**	Global generic design
Data Security	**AES-256 + TLS 1.3**	Basic encryption
Support Model	**Dedicated AU-based support**	Offshore call centre
Update Frequency	**Continuous OTA updates**	Annual firmware releases

9. Integration & Ecosystem

Secure Vault integrates seamlessly with the broader Palatial Pools technology ecosystem through standardised APIs, shared data models, and the centralised Gods Eye Command Centre.

• **Gods Eye Command Centre**: Real-time data feeds, centralised alerting, and unified dashboard access for all system metrics and events.

• **Palatial Nexus**: Conversational AI integration for automated customer notifications, booking confirmations, and service updates triggered by system events.

• **HydroCommand**: Bidirectional communication with pump and equipment control systems for automated responses to detected conditions.

• **Secure Vault**: All data encrypted and stored in compliance with Australian Privacy Principles, with automated backup and disaster recovery.

• **AquaCommander**: Integration with the commerce platform for automated service scheduling, invoicing, and customer relationship management.

10. Security & Compliance

All data processed by Secure Vault is protected by enterprise-grade security measures, including AES-256 encryption at rest, TLS 1.3 encryption in transit, and role-based access control (RBAC). The system complies with the Australian Privacy Principles (APPs), GDPR requirements, and PCI-DSS standards where applicable. Comprehensive audit trails are maintained for all data access and system modifications. Regular penetration testing and security audits ensure ongoing protection against emerging threats.

11. Contact & Next Steps

Palatial Pool Services Pty Ltd
Technology Division

Melbourne & Queensland, Australia

Phone: 0452 157 694

Email: service@palatialpools.com.au

Web: www.palatialpools.com.au

Schedule a Technology Demonstration

Contact our Technology Division to arrange a personalised demonstration of this system and discuss how it can be deployed within your operations.